



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520030492
Branch	CSE
Course Outcome	CO1

Case Study Topic: Recursive DNS Resolver – TTL Driven Cache Expiry using Min Heap and HashMap

Work Done Summary:

In this case study, a DNS cache management system was implemented using a Min Heap and HashMap. The system stores domain records along with their expiration times (TTL) and efficiently removes expired entries without scanning the entire cache.

Implementation Details:

- Created a CacheEntry class to store domain name, expiration time, and DNS record.
- Implemented a Min Heap (Priority Queue) to maintain entries based on expiration time.
- Used a HashMap for fast domain lookup operations.
- Inserted DNS records into both Min Heap and HashMap.
- Implemented cache eviction using heap root checking.
- Removed expired records from both data structures.
- Handled TTL renewal using updated cache entries.
- Displayed evicted domains after expiration processing.

Result: Screenshots/Output

```
DNS CACHE EXPIRY SYSTEM
-----
Current Time : 1700000030

Expired Domains:
kl.ac.in
hdfc.in

Total Expired Domains = 2

=== Code Execution Successful ===
```

Observations:

DNS records were successfully inserted into the cache.

- Domain lookup was performed efficiently using HashMap.
- Expired entries were removed correctly using Min Heap operations.
- Cache eviction was faster than periodic full-cache scanning.
- The Min Heap always provided the next entry to expire.
- The implementation reduced cache maintenance overhead.

GitHub Link: <https://github.com/eswarikalipriya-boop/DSA.git>

Student Signature	Faculty Signature